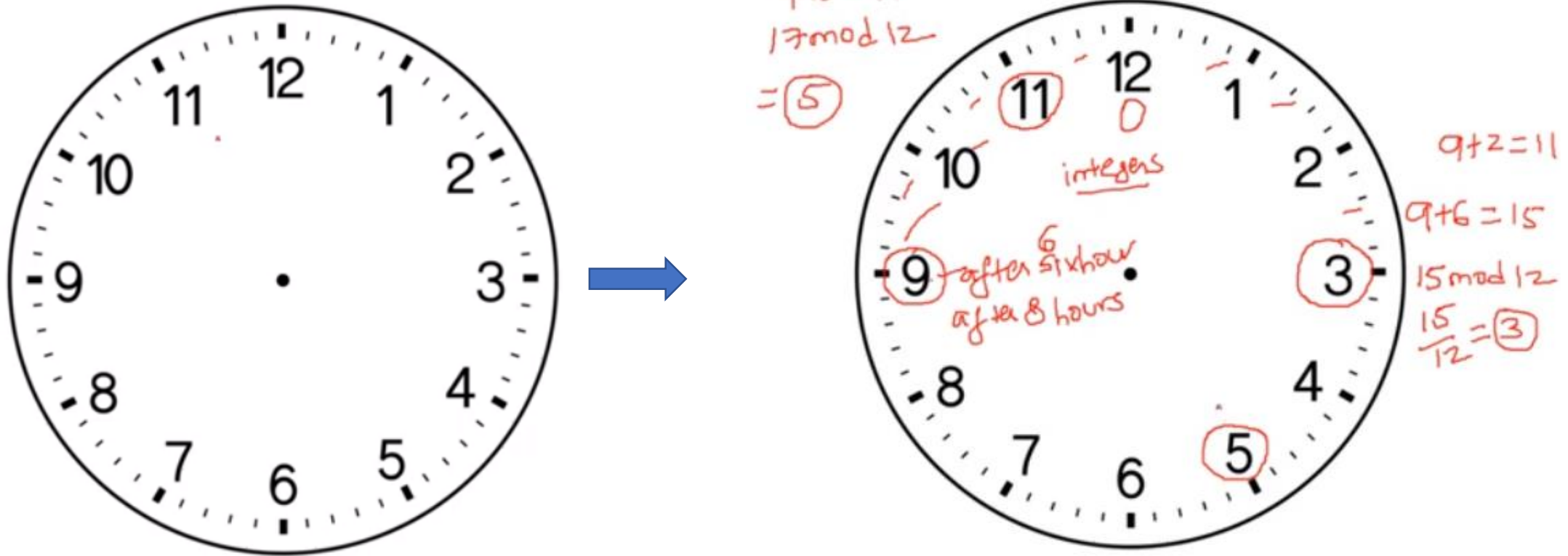


Modular Arithmetics

Module I

Modular Arithmetic's

- It is an arithmetic for Integers that focus on the remainder.
- Numbers wrap around on reaching a certain value.

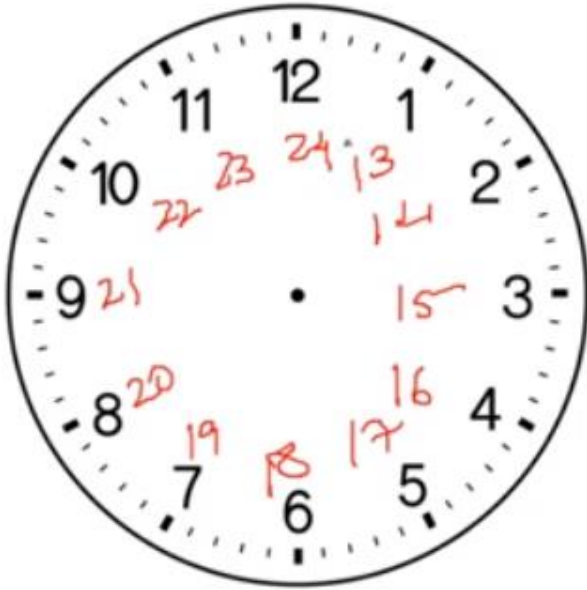


Exercise

- $18 \bmod 12 \rightarrow 18/12 = 6$
- $24 \bmod 12 \rightarrow 24/12 = 0$
- $32 \bmod 12 \rightarrow 32/12 = 8$
- $4 \bmod 12 \rightarrow 4$
- $8 \bmod 12 \rightarrow 8$ (How many 12's in 8 is 0 (remainder))

Here, 12 is called as modulus.

Congruence



- Here, 3 is equals to 15 with respect to 12 hour clock.
- $15 \equiv 3 \pmod{12}$, 15 is congruent to 3 w.r.t 12 hour clock.
- 6 is equals to 18 with respect to 12 hour clock.
- $18 \equiv 6 \pmod{12}$, 18 is congruent to 6 w.r.t 12 hour clock.

Congruence

- If a and b are Integers and $n > 0$, we can write $a \equiv b \pmod{n}$ to mean $n \mid (a-b)$.

Example: $15 \equiv 3 \pmod{12} \rightarrow 12 \mid (15-3) \rightarrow 12 \mid 12$ proved.

Properties of congruence

- $a \equiv r \pmod{n}$, if r is the remainder obtained by dividing a by n .

Proof: $15 \equiv 3 \pmod{12} \rightarrow 12 \mid 15-3 \rightarrow 12 \mid 12, n \mid a-r, n \text{ divides } a-r.$

- $a \equiv a \pmod{n} \rightarrow$ reflexivity.

Proof: $n \mid a-a \rightarrow n \mid 0$

- $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n} \rightarrow$ symmetry.

Proof: from the simple property of a divisor,

$$n \mid a \text{ and } n \mid b \rightarrow n \mid a + b$$

$$5 \mid 20 \text{ and } 5 \mid 10 \rightarrow 5 \mid (20+10) \rightarrow 5 \mid 30.$$

$$n \mid (a-b) \quad \text{and} \quad n \mid (b-a) \quad \rightarrow \quad n \mid a - b + b - a \rightarrow n \mid 0$$

- $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$. $\rightarrow$ Transitivity.

if $n \mid a-b$ and $n \mid b-c$ then $n \mid a-b + b - c \rightarrow n \mid a-c \rightarrow a \equiv c \pmod{n}$

Properties of congruence

Addition by a constant

- If $a \equiv b \pmod{n}$ and $c \neq 0$ then
- $a + c \equiv b + c \pmod{n}$
- $a - c \equiv b - c \pmod{n}$
- $a + c \equiv b + c \pmod{n + c}$ **is not valid.**

E.g. $15 \equiv 3 \pmod{12}$, let $c=2$

$$15+2 \equiv 3+2 \pmod{12} \rightarrow 17 \equiv 5 \pmod{12} \rightarrow 12 \mid 17-5 \rightarrow 12 \mid 12 .$$

$$15-2 \equiv 3-2 \pmod{12} \rightarrow 13 \equiv 1 \pmod{12} \rightarrow 12 \mid 13-1 \rightarrow 12 \mid 12 .$$

Properties of congruence

- *Multiplication by a constant*
- If $a \equiv b \pmod{n}$ and $c \neq 0$ then
- $a * c \equiv b * c \pmod{n}$
- $a * c \equiv b * c \pmod{n * c}$ **is valid.**

E.g. $15 \equiv 3 \pmod{12}$, let $c=2$

$$15 * 2 \equiv 3 * 2 \pmod{12}$$

$$30 \equiv 6 \pmod{12}$$

$$12 \mid 30 - 6 \rightarrow 12 \mid 24 \text{ True}$$

or

$$30 \bmod 12 \equiv 6 \bmod 12 \rightarrow 6 \equiv 6$$

E.g. $15 \equiv 3 \pmod{12}$, let $c=2$

$$15 * 2 \equiv 3 * 2 \pmod{12 * 2}$$

$$30 \equiv 6 \pmod{24}$$

$$24 \mid 30 - 6 \rightarrow 24 \mid 24 \text{ True}$$

or

$$30 \bmod 24 \equiv 6 \bmod 24 \rightarrow 6 \equiv 6$$

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Modulo of a Negative Number

- $3 \bmod 12$? Ans: 3
- $7 \bmod 12$? Ans: 7
- $13 \bmod 12$? Ans: 1
- $-3 \bmod 12$? Ans: 9 (Anti-clockwise)
- $-7 \bmod 12$? Ans: 5 (Anti-clockwise)
- $-13 \bmod 12$? Ans: 11 (Anti-clockwise)

General formula:

$$-n \bmod k = k - (n \bmod k)$$

- $-3 \bmod 12$? $12 - (3 \bmod 12) = 12 - 3 = 9$
- $-7 \bmod 12$? $12 - (7 \bmod 12) = 12 - 7 = 5$
- $-13 \bmod 12$? $12 - (13 \bmod 12) = 12 - 1 = 11$
- $-3 \bmod 20$? $20 - (3 \bmod 20) = 20 - 3 = 17$

